

REPORT 06 OF 6 — READ THIS ONE FIRST

Consolidated Overview & Developer Handoff

The map of the whole Ignition package: what the five specialist reports decided, the numbers they commit to, one merged implementation roadmap across backend, web, WPF and mobile, QA acceptance checklists, the risk register, and the short list of decisions that still belong to Ralph.

Package Index

Decision Register

Unified Roadmap

Acceptance Criteria

Risk Register

Owner Questions



Table of Contents

01	Executive Summary	What the Ignition package is, the two screens, the headline numbers	p. 03
02	Package Map	How the six reports interlock — dependency figure and reading order	p. 04
03	Package Index — Report 01	UX Research & User Flows (dev-ui-ux)	p. 05
	Package Index — Report 02	Visual Design Language (dev-graphic-designer)	p. 06
	Package Index — Report 03	UI Component Architecture (dev-web-react)	p. 07
	Package Index — Report 04	System Architecture (dev-backend-api)	p. 08
	Package Index — Report 05	Workflow Optimization (workflow agent)	p. 09
04	Unified Decision Register	The cross-cutting decisions all five reports agree on	p. 10
05	Consolidated Implementation Roadmap	One sequenced plan: Phase 0–3, owners, files, endpoints, exit criteria	p. 11–12
06	Acceptance Criteria — Dashboard	Testable QA checklist: visual, functional, performance, a11y, security	p. 13
	Acceptance Criteria — Client Workspace & Cross-Cutting	Workspace checklist plus package-wide budgets and security gates	p. 14
07	Risk Register	Top 10 risks with likelihood, impact and mitigation	p. 15
08	Open Questions for Ralph	Decisions the package deliberately leaves to the owner	p. 16

Figures in this report: Figure 1 — Package map (six reports and their dependencies), p.04. Figure 2 — Consolidated roadmap timeline (Phase 0–3 with approval gates), p.11.

How to use this report. A developer new to the package reads pages 3–4 for orientation, jumps to the report index page for their discipline (pages 5–9), then works from the roadmap (pages 11–12). QA works from pages 13–14. Nothing in this report introduces a new decision — every figure and rule here is traceable to one of the five source reports, which remain the authoritative specifications.

Executive Summary

A from-scratch redesign of the two screens where the business actually lives — grounded in the current codebase, quantified, and ready to build.

What this package is. "Ignition" is a complete, ground-up redesign of the SpareParts **Dashboard** and a new unified **Client Workspace**, produced by a six-agent design team as six companion reports. It replaces the rejected "Aurora" navy/teal direction with a deliberately unmistakable identity — carbon black surfaces (#0a0a0b / #131316 / #1b1b1f), a single electric-orange accent (#f97316), Plus Jakarta Sans / Inter / JetBrains Mono type, borderless shadow cards at 16px radius, a 72px icon rail, and a global `ctrl1 + k` command bar. Every claim in the package is grounded in real files: `screen-registry.js` (70 registered screens), `dashboard-view.js` (663 lines, 7 API calls on mount), the 17,400-line `styles.css`, and the existing `OwnerCockpitService` aggregate.

The two screens. (1) The Dashboard becomes a **command center**: a KPI band of tabular-numeral tiles, a 7-day trend, and — promoted to the organizing principle of the screen — the **Action Queue**, which scores exceptions by severity × money at stake, carries a lifecycle (appears → acted → resolved/dismissed), and deep-links to the exact entity, not just a screen. (2) The **Client Workspace** replaces six fragmented screens (Contacts, Customer Aging, Invoices, Quotes, Part Requests, WhatsApp) with one customer-360 record — identity, balance and aging buckets, invoices, payments, quotes, requests and communications as tabs of a single view, where a payment can be received and a reminder previewed and sent without navigating away.

799 → 358

UI steps per shop-day across the 10 highest-frequency tasks (~55%) — Report 05

~98 min

Projected daily UI-tax recovery per shop (~1.6 h); verified by telemetry before legacy retirement — Report 05

95s → 30s

Two-line counter sale, keyboard-first POS loop + Ctrl+K grammar — Report 05

7 → 1

Dashboard API requests: one composed GET `/api/dashboard/summary` — Report 04

≤350 ms

p95 server budget, dashboard cache-miss; ≤12 ms on ETag 304 cache-hit — Report 04

6 → 1

Screens to answer "what does this client owe and why?" — Reports 01/05

68 / 70

Legacy screens untouched: Ignition ships scoped under `.ignition-root`, zero regression — Report 03

WCAG 2.2 AA

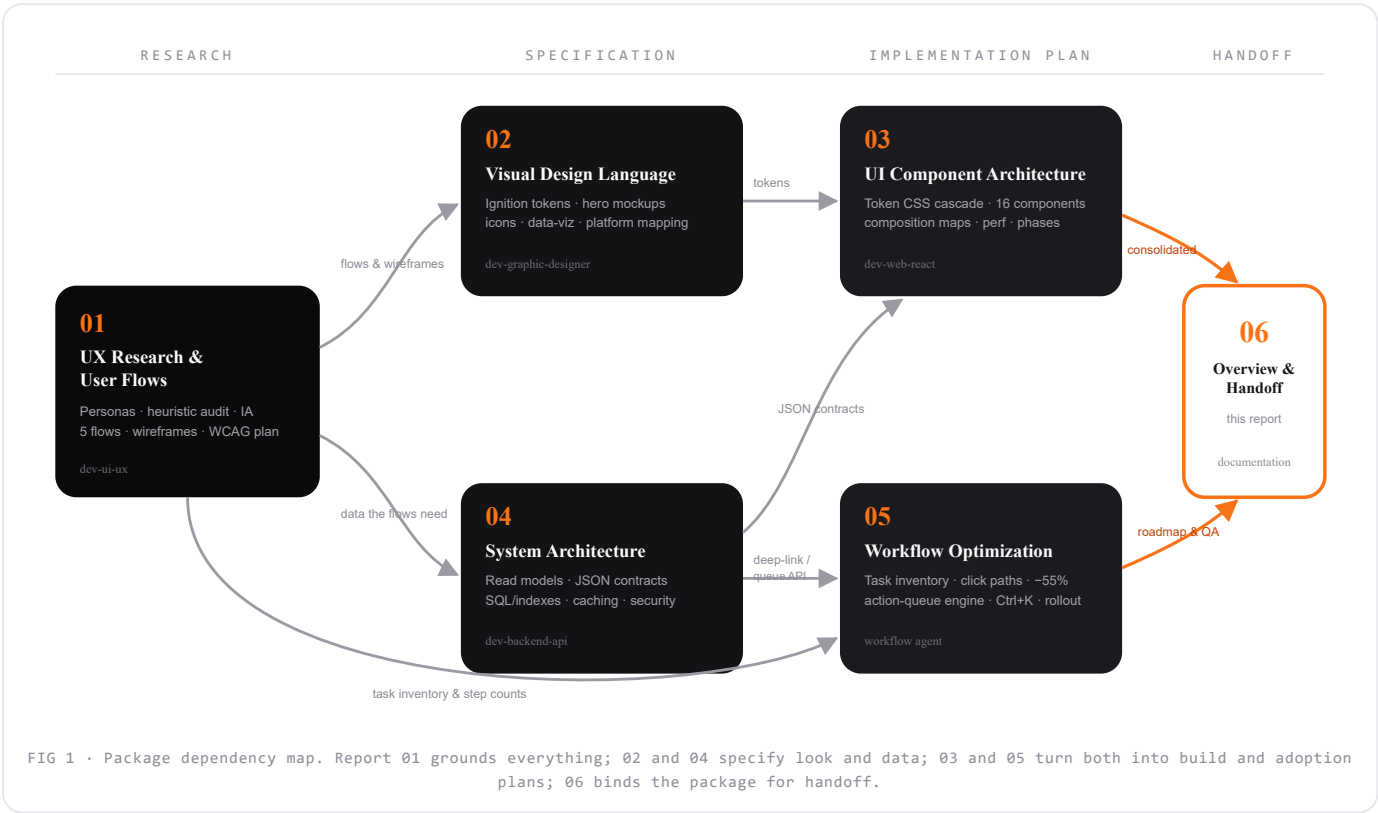
Contrast-verified token pairs (19.4:1 body, 7.6:1 orange CTA), keyboard map, reduced motion — Reports 01/03

How it ships. Everything is additive. The backend adds 8 new endpoints and changes none (`/api/owner-cockpit` and friends keep working for WPF and mobile until each migrates on its own schedule). The web app adds a four-layer token CSS cascade after `styles.css` and two rebuilt views; the other 68 screens render exactly as today. Rollout is side-by-side for four weeks with telemetry gates — legacy screens retire only when measured targets are hit (counter sale p50 ≤35s, queue dismissals ≤15%), never on a calendar date, and file deletions require Ralph's explicit approval per the workspace safety rules.

Bottom line for the dev team: two screens, ~24 new frontend files, 8 additive endpoints, 4 covering indexes, one token contract shared by web, WPF and mobile — with a measured payback of roughly one working month of staff time per shop per year if the projections hold, and instrumentation to prove whether they do.

Package Map — How the Six Reports Interlock

Research feeds specification; specification feeds implementation; this report is the single entry point that binds them.



Recommended reading order per role

ROLE / AGENT	READ FIRST	THEN	REFERENCE AS NEEDED
dev-web-react	06 (this) → 03 all sections	02 §13 token mapping · 04 §02–06 contracts	01 wireframes · 05 §4–6 queue/palette behavior
dev-backend-api	06 → 04 all sections	05 §4 deep-link contract · 01 §01 missing endpoints	03 §6–7 composition maps (what binds to what)
dev-database	06 → 04 §07 SQL & indexes	04 §04 caching/invalidation	04 §09 tenant-guard rules
dev-desktop-wpf / dev-mobile-rn	06 → 02 §13 token mapping	01 §12/§14 narrow wireframes · 05 §7 cross-device	04 §11 additive versioning (no breakage)
dev-qa-security	06 §06–07 (pages 13–15)	04 §09 security · 01 §15 WCAG plan	03 §10–11 budgets · 05 §8 telemetry gates
Ralph (owner)	06 §01 + §08	02 hero mockups (pages 8–12)	05 §1 + Fig 6 (the payoff)

01

UX Research & User Flows

docs/redesign/01-ux-research-user-flows.html · 22 pages

AUTHOR

dev-ui-ux — UX Research Agent

PURPOSE

Diagnose today's UX in the real code; define the IA, flows and wireframes everything else builds on

GROUNDING IN

dashboard-view.js · contacts-view.js · customer-aging-view.js · screen-registry.js

Section list

1. Executive Summary of UX Findings

2. Personas & Jobs-To-Be-Done (Rita owner · Karim counter · Nour accounting · Elie warehouse)

3. Heuristic Audit — Dashboard & Client/Contacts (Top 10 Problems)

4. New Information Architecture — 70+ Screens

5. IA Map — Icon Rail, Command Palette, Progressive Disclosure (Figure)

6. Flow 01 — Morning Business Review

7. Flow 02 — Invoicing a Walk-in Client

8. Flow 03 — Client Lookup Mid-Counter-Sale
9. Flow 04 — Chasing Overdue Receivables

10. Flow 05 — Quote → Invoice Conversion

11. Wireframe — Dashboard (Desktop)

12. Wireframe — Dashboard (Mobile / Narrow)

13. Wireframe — Client Workspace (Desktop)

14. Wireframe — Client Workspace (Mobile / Narrow)

15. Accessibility Compliance Plan — WCAG 2.2 AA

16. Cognitive Load & Progressive Disclosure Principles

Most important decisions

- 1.1

Triage-first dashboard. Lead the screen with the exceptions `buildActionQueue()` already computes (negative P&L, unpaid receivables, red stock, failed messages) instead of burying them under 20+ undifferentiated data blocks.
- 1.2

One Client Workspace replaces six screens. Contacts, Customer Aging, Invoices, Quotes, Part Requests and WhatsApp become tabs of a single client record — and the report explicitly flags the missing backend dependency: no `GET /api/customers/{id}` exists today (handed to Report 04).
- 1.3

New IA: 72px icon rail + Ctrl+K palette + progressive disclosure. Replaces the flat 7-group sidebar whose "More Modules" bucket silently absorbs 40+ of the 70 registered screens.
- 1.4

Audited flows with committed step counts. Morning review 6→2 · walk-in invoice 7→4 · mid-sale client lookup 8→2 · overdue chase (3 clients) 9→4 · quote→invoice 8→4 — a 35–60% step reduction with no data hidden.
- 1.5

WCAG 2.2 AA plan is binding on every wireframe. Contrast strategy, keyboard navigation map, focus order, screen-reader landmarks, reduced motion, and 44px touch targets (§15) plus two shop-floor rules: most-urgent action reachable without scrolling; re-typing a customer's name in a second screen is a defect.

02 Visual Design Language

docs/redesign/02-visual-design-system.html · 15 pages — the approval artifact for Ralph's mockup sign-off

AUTHOR

dev-graphic-designer — Visual Design Agent

PURPOSE

Define the Ignition identity: tokens, ramps, type, elevation, hero mockups, icons, chart style, platform mapping

DESIGN TEST

"If it isn't carbon black with one orange accent and machined type, it isn't Ignition"

Section list

1. Design Vision — Ignition (why carbon + orange)

2. Color System — Dark Mode (default surface)

3. Color System — Light Mode

4. Typography, Spacing & Radii (4/8 grid)

5. Elevation & Motion (4 shadow levels, duration/easing)

6. Before → After (Apex/Aurora vs Ignition)

7. Hero Figure A — Dashboard (Desktop)
8. Hero Figure B — Client Workspace (Desktop)

9. Mobile Variants (narrow layout)

10. Light-Mode Dashboard Variant

11. Iconography (stroke spec + 12-icon set)

12. Data Visualization Style

13. Cross-Platform Token Mapping (CSS → WPF → RN)

Most important decisions

- 2.1

Carbon + one orange accent. Three-step surface elevation (#0a0a0b → #131316 → #1b1b1f) (+ #232328 pressed), accent #f97316 (hover #ea580c) used sparingly enough that orange always means "look here", four semantic status colors (#22c55e / #fbbf24 / #ef4444 / #38bdf8). Full 50–900 ramps for neutral and orange; complete light-mode inversion included.
- 2.2

Three-typeface stack with strict roles. Plus Jakarta Sans (display, −0.04em tracking), Inter (body — disappears at small sizes in dense tables), JetBrains Mono reserved exclusively for tabular numerals: prices, quantities, aging buckets.
- 2.3

Borderless shadow cards, 16px radius, dark-first. Cards separate by layered shadow and one-shade surface shifts, never borders; geometry gets sharp only for status (pills, 2px orange active-nav rail).
- 2.4

Single stroke icon family. 24×24 grid (20×20 live area), 1.75px stroke, round caps/joins, stroke-only with accent fill reserved for active state; optical sizes 14/16/20/28px.
- 2.5

One token system, three implementations. The §13 table is the cross-platform contract: CSS --ig-* custom properties ↔ WPF IgnitionTheme.xaml ResourceDictionary keys (IgBrushAccent ...) ↔ React Native theme.js. 72px rail on desktop/web; bottom tabs + 44px targets are mobile's only sanctioned deviations.

Workflow gate: per the Design Change Workflow in CLAUDE.md, the hero mockups in this report (pages 8–12) are the "before pictures" Ralph must approve before any implementation phase starts, and post-implementation screenshots must be compared back to them.

03 UI Component Architecture & Patterns

docs/redesign/03-ui-component-architecture.html · 14 pages

AUTHOR

dev-web-react — Web Frontend Agent

PURPOSE

Turn Reports 01+02 into buildable components inside the no-build hyperscript React runtime

HARD CONSTRAINTS

No JSX · no bundler · plain ES modules with `h()` · one component per file · ApiClient unchanged

Section list

1. Executive Summary (state of the component layer)
2. Design-Token Architecture
3. New Component Inventory — Shell & Data
4. New Component Inventory — Client Workspace
5. Component States & Specimen Renders
6. Composition Map — Dashboard Screen
7. Composition Map — Client Workspace Screen
8. Responsive Strategy (desktop / tablet / phone)
9. Interaction Engineering (focus, skeletons, timing)
10. Performance Budget
11. Accessibility Implementation
12. Implementation Plan & File Map (Phase 0–3)

Most important decisions

- 3.1 Four-layer CSS cascade, additive to the legacy file.** `tokens.css` → `primitives.css` → `components.css` → `views.*.css` linked *after* the 17,400-line `styles.css`; every rule scoped under an opt-in `.ignition-root` class so the 68 untouched screens can't regress. Total new CSS <25KB gzipped.
- 3.2 Second-generation component set under `js/components/ignition/`.** 16 new components — IconRail, CommandBar, CommandPalette, KpiTile, KpiBand, ActionQueueCard, InsightPanel, TrendChart, DataGrid, MoneyText, QuickActionDock, ClientRail, ClientHeader, AgingBar, ActivityTimeline, TabStrip — composing over (never replacing in place) the existing `ui/` kit; chart math and aging-bucket math ported from the live views.
- 3.3 Performance budget is contractual.** LCP <1.8s (KpiBand is the LCP element), Total Blocking Time <150ms, skeleton→content crossfade 160ms with a ≤150ms skip rule to avoid loading-state flash, palette debounce 180ms.
- 3.4 Responsive contract per breakpoint.** 12-col desktop → 8-col tablet (bottom tab bar, KPI 2-col scroll-snap) → 4-col phone (bottom tabs + "More" sheet, KPI carousel, ClientRail becomes a full-screen drawer with back-button paging).
- 3.5 Phased, reversible file map.** Phase 0 tokens → Phase 1 component library → Phase 2 Dashboard v2 (same `load()` and `buildActionQueue()` logic retained) → Phase 3 Client Workspace. Explicit do-not-touch list (`styles.css`, `api-client.js`, all 66 other views); superseded views stay on disk pending Ralph's deletion approval.

04 System Architecture — Read Models & Contracts

docs/redesign/04-system-architecture.html · 13 pages

AUTHOR

dev-backend-api — Backend API Agent

PURPOSE

The composed endpoints, JSON contracts, SQL, caching and security the two screens bind to

POSTURE

Everything additive — zero changes to any currently shipping endpoint; WPF/mobile unaffected

Section list

1. Executive Summary (chatty-call inventory)

2. Dashboard Read Model — GET /api/dashboard/summary

3. Dashboard — Example Payload

4. Caching & Invalidation

5. Client Workspace Read Model — GET /api/clients/{id}/workspace

6. Workspace Child Resources (paged tabs + typeahead)
7. Data Layer — Aggregate SQL & Covering Indexes

8. Performance Engineering (N+1, compression, ETag)

9. Security Architecture

10. Sequence Diagrams

11. Versioning & Rollout

Most important decisions

- 4.1

One composed dashboard endpoint. GET /api/dashboard/summary wraps the existing OwnerCockpitService (unchanged) plus new DashboardActionQueueService and DashboardTrendService, collapsing 7 client calls to 1. Budget: ≤350ms p95 server time cold, ≤12ms on ETag 304; core P&L stays in one DbSession for consistency, independent reads parallelize via Task.WhenAll on separate short-lived connections.
- 4.2

The Client Workspace becomes a backend resource. GET /api/clients/{id}/workspace (header + balance + aging + KPIs, 20s TTL) with five independently paged child routes — /invoices /payments /quotes /part-requests /timeline — plus /api/clients/search typeahead. 8 new endpoints total.
- 4.3

Cache + ETag polling, not SignalR. Per-tenant cache keys (dash:tenant:{id}:{date} , 45s TTL) with explicit write-path invalidation on sales/payments so money is never stale; dashboard "Live updates" polls at 30–60s with If-None-Match . SignalR rejected for cost and cross-tenant-broadcast risk.
- 4.4

Covering indexes ship first. Additive migration (e.g. IX_Transactions_Tenant_Type_Date INCLUDE(...)) deployable ahead of code; every query keeps the codebase's tenant-guard shape (@TenantId = 0 OR t.TenantId = @TenantId) with TenantId sourced only from the JWT via ITenantContext — never from a request.
- 4.5

Security invariants. Class-level [Authorize] on both new controllers; uniform 404 for missing or cross-tenant customers (enumeration prevention); zero tenantId / dbName /cache-key leakage in wire DTOs; production-safe ProblemDetails errors; search rate-limiting partitioned by tenant, not IP.

05 Workflow Optimization

docs/redesign/05-workflow-optimization.html · 15 pages

AUTHOR

Workflow Optimization Agent

PURPOSE

Quantify the redesign's payback task-by-task and define the adoption, telemetry and rollback plan

DISCIPLINE

"No projection ships unmeasured" — every \$3 number becomes a telemetry distribution

Section list

1. Executive Summary (where time is lost today)

2. Task Inventory (T1–T10, roles, frequencies)

3. Click-Path Analysis — Before vs After

4. The Action Queue as a Workflow Engine

5. Client Workspace — Operational Flows
6. Speed Mechanics (Ctrl+K grammar, barcode, recents)

7. Cross-Device Workflow (phone vs desktop, offline)

8. Adoption & Rollout (telemetry, gates, rollback)

9. Figures F1–F6 (swimlanes, lifecycle, day-in-the-life)

Most important decisions

- 5.1 **The headline numbers.** Across the 10 highest-frequency tasks: **799 → 358 UI steps per shop-day (–55%)**, projected **~98 min/day** recovered (~47 min counter staff, ~43 min owner, ~8 min quoting — ~1.6h, one working month/year); 2-line counter sale **95s → ~30s**. Fig 6's whole-day model shows the UI tax falling 98 → 41 min (–58%). Assumed frequencies are replaced by measured counts during rollout.
- 5.2 **Action Queue as a workflow engine, not a hint list.** Items scored by severity × money at stake, full lifecycle (appears → acted → resolved/dismissed) that survives refresh, and a deep-link contract into the exact entity-level action — upgrading the existing `buildActionQueue()` six signals.
- 5.3 **Ctrl+K grammar.** Terse verbs: `inv eli` (invoice for Elie), `pay eli 250`, `part n54 coil` (inline stock/price card — no navigation), `q eli`, `c eli`, `cust +`, `wa eli`, `#8811`, `> screen`. <50ms keystroke-to-result on cached index; barcode scanners act as keyboard wedges into the same field.
- 5.4 **Missing flow discovered and fixed: receiving a payment.** Today it has no UI path at all (manual journal in Accounting); the Workspace payment sheet makes it a 3-step flow with oldest-first allocation (T4 target: 3min → ≤45s).
- 5.5 **Metric-gated rollout, not calendar-gated.** 4-week side-by-side (opt-in weeks 1–2, default weeks 3–4), 6 telemetry event families, 9 success metrics (e.g. T2 p50 ≤35s, queue same-day resolution ≥70%, dismissals ≤15%), and hard rollback criteria (T2 p50 >60s or dismissals >30% at week 4 flips the default back).

Count note for implementers: Report 05 counts 74 registry entries where Reports 01/03 count 70 — the registry grew during the package's production. Treat `screen-registry.js` at build time as the source of truth; the "68 untouched screens" guarantee means "all registered screens except the two Ignition rebuilds".

Unified Decision Register

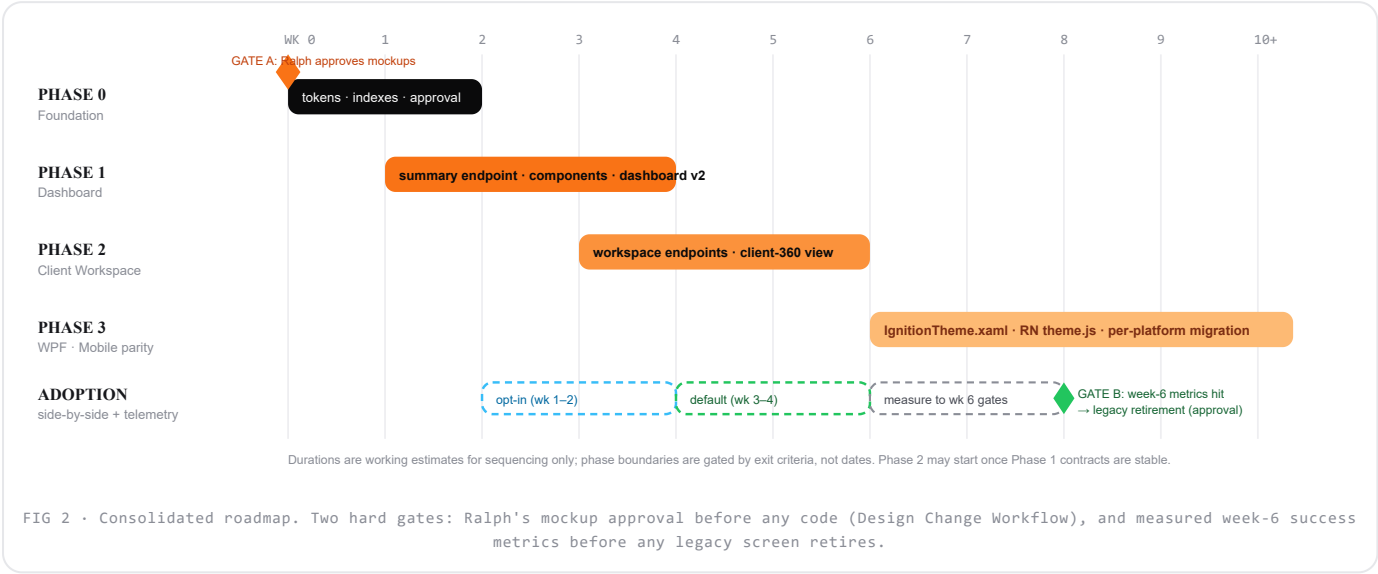
The cross-cutting decisions every report agrees on. If an implementation choice contradicts a row below, stop and escalate — these are the package's load-bearing walls.

#	DECISION	RATIONALE (ONE LINE)	SPECIFIED IN DETAIL
D1	Ignition token system — carbon 3-step elevation + single orange accent + 3-typeface stack; dark-first with full light mode	Visibly, unmistakably different from Aurora and the live graphite-blue theme; orange = "look here", used for exactly one thing per screen	02 §1–5 (spec) · 03 §2 (CSS encoding) · 02 §13 (WPF/RN mapping)
D2	Action Queue is the dashboard's organizing principle — scored, lifecycled, deep-linked	The triage list the code already computes is the highest-value pixel on the screen; charts support it, not vice versa	01 §1/§6 (why) · 04 §2 (server-side build) · 05 §4 (engine + lifecycle contract)
D3	Client-360 model — one workspace per client; six fragmented screens become tabs of one record	Re-searching the same customer across 6 screens is the single largest owner-side time sink	01 §1/§13 (IA + wireframe) · 03 §4/§7 (components) · 04 §5–6 (contract) · 05 §5 (flows)
D4	Single composed dashboard endpoint — <code>GET /api/dashboard/summary</code> , 7 calls → 1	Over-fetch (full parts catalog, full aging table) is a server-side aggregation problem, not a frontend caching problem	04 §2–4 (contract, budget, cache) · 03 §6 (binding)
D5	Ctrl+K command palette on every screen — terse grammar, inline answers, barcode wedge	Counter tasks run ~55×/day; two keystrokes beat any menu, and the 70-screen long tail stays reachable	01 §4.1 · 03 §3 (component) · 05 §6 (grammar + latency)
D6	WCAG 2.2 AA — contrast-verified tokens, keyboard map, focus management, reduced motion, touch targets	Shop-floor use means bad light, gloves, hurry — accessibility here is usability for everyone	01 §15 (plan) · 02 §2–3 (AA-calibrated ramps) · 03 §11 (ARIA/impl)
D7	Additive, phased shipping with zero legacy regression — scoped CSS, additive endpoints, side-by-side rollout	Two screens must ship without risking the other 68 or breaking WPF/mobile clients that still call old routes	03 §2/§12 (scoping + file map) · 04 §11 (versioning) · 05 §8 (adoption gates)
D8	Tenant-isolation invariants — TenantId from JWT only; uniform 404; no tenant/dbName in responses	Multi-tenant money data: enumeration and leakage prevention are non-negotiable and already codebase convention	04 §5/§7/§9 · CLAUDE.md security focus
D9	ETag polling over SignalR — 30–60s poll, 304s near-free	A parts-shop dashboard doesn't need sub-second push; polling is simpler, cheaper, and safer for tenant isolation	04 §8 (comparison table)
D10	Measure before retiring anything — telemetry gates, rollback criteria, deletions need Ralph's approval	The projections (~55%, ~98 min) are planning numbers until the beacons confirm them; honesty is designed in	05 §8 (events, gates, rollback) · 03 §12 (files kept on disk)

Change control: Report 01 already states it — any deviation from the agreed IA (and by extension any D-row above) must be re-approved by Ralph before implementation, per the Design Change Workflow. The register exists so a reviewer can check a PR against ten rows instead of five reports.

Consolidated Implementation Roadmap

Report 03's build phases, Report 04's endpoint rollout and Report 05's adoption plan merged into one sequence. Owner column = dev team agents from the workspace roster.



PHASE 0 Foundation — tokens, indexes, owner approval ~1 week

OWNER	SCOPE & FILES/ENDPOINTS	EXIT CRITERIA
dev-ui-ux dev-graphic-designer	Present Report 02 hero mockups (dark + light, both screens, mobile variants) to Ralph for sign-off — Gate A , required by the Design Change Workflow before any implementation.	Written "yes" from Ralph on the Ignition direction; chosen default theme recorded (see §08 Q1).
dev-web-react	Create <code>wwwroot/css/ignition/tokens.css</code> + <code>primitives.css</code> (all <code>--ig-*</code> props, dark + <code>[data-ig-theme="light"]</code>), scoped <code>.ignition-root</code> ; edit <code>index.html</code> to link them after <code>styles.css</code> .	App builds & loads; pixel-identical rendering of all legacy screens (spot-check 10); new CSS ≤6KB.
dev-database	Migration with covering indexes from Report 04 §07 (<code>IX_Transactions_Tenant_Type_Date</code> et al). Additive only. Requires explicit approval per CLAUDE.md before running on any shared/production DB; low-traffic window.	Indexes present; query plans for aging/trend/summary show index use; zero data change.
dev-qa-security	Baseline capture: current screen renders, current p95 latencies of the 7 dashboard calls, week-0 overdue-receivables balance (Report 05 metric baseline).	Baseline document stored with the package.

Roadmap — Phases 1 to 3

PHASE 1 Dashboard — endpoint, component library, first shipped screen

~3 weeks

OWNER	SCOPE & FILES/ENDPOINTS	EXIT CRITERIA
dev-backend-api	DashboardController + DashboardActionQueueService + DashboardTrendService → GET /api/dashboard/summary ; per-tenant cache (45s TTL) + weak ETag; then IDashboardCacheInvalidator hooks on sale/payment commit paths (the one medium-risk step — full build+test after).	Contract matches Report 04 §03 payload; p95 ≤350ms cold / ≤12ms 304 under test load; invalidation proven by integration test (post payment → next poll shows new KPI).
dev-web-react	components.css + 11 shell/data components (IconRail, CommandBar, CommandPalette, KpiTile/Band, ActionQueueCard, InsightPanel, TrendChart, DataGrid, MoneyText, QuickActionDock); rebuild dashboard-view.js on the new endpoint wrapped in .ignition-root ; palette grammar verbs inv/pay/part/q/c/cust/wa/#/> ; telemetry beacons (Report 05 §8 events).	Dashboard renders from 1 composed call (+2 reference-data calls); LCP <1.8s, TBT <150ms; palette <50ms on cached index; screenshots sent to Ralph vs Report 02 Fig A (Design Change Workflow step 2).
dev-qa-security	Contract tests, a11y audit (keyboard map, contrast, focus return), security checks (401 unauthenticated, no tenant/dbName in payload, ProblemDetails shape), perf regression run.	Page-13 checklist fully green; opt-in banner live (adoption weeks 1–2 start).

PHASE 2 Client Workspace — client-360 endpoints and screen

~3 weeks, overlaps P1 tail

OWNER	SCOPE & FILES/ENDPOINTS	EXIT CRITERIA
dev-backend-api	ClientWorkspaceController/Service : GET /api/clients/{id}/workspace (20s TTL) + paged /invoices /payments /quotes /part-requests /timeline + /api/clients/search ; explicit tenant-ownership check before child queries; uniform 404.	All 7 routes live & paged via existing pagination headers; enumeration test passes (cross-tenant id → identical 404).
dev-database	Single-customer aging query (narrowed CTE from CustomersService.GetAging), timeline union query, parameterization review of all new SQL.	No string-concatenated SQL; plans use Phase-0 indexes.
dev-web-react	5 workspace components (ClientRail, ClientHeader, AgingBar, ActivityTimeline, TabStrip) + client-workspace-view.js ; payment sheet, invoice/quote-from-context, reminder preview (reuses CommunicationPayloadFactory.accountBalanceReminder); registry + Sidebar "Clients" entry; legacy contacts/aging keys stay registered.	All Report 05 §5 flows work end-to-end; T4 receive-payment flow ≤45s in hallway test; after-screenshots to Ralph vs Report 02 Fig B; default flip (adoption weeks 3–4).
dev-qa-security	Page-14 checklist; load test workspace + child tabs; verify telemetry event flow for T2/T4/T6.	Checklist green; week-6 metric collection running toward Gate B.

PHASE 3 WPF & mobile parity, hardening, legacy retirement

~4 weeks, own pace per platform

OWNER	SCOPE & FILES/ENDPOINTS	EXIT CRITERIA
dev-desktop-wpf	IgnitionTheme.xaml ResourceDictionary (1:1 IgBrush*/IgFont*/IgCornerRadius* keys per Report 02 §13) merged in App.xaml ; Dashboard + Client Workspace views consume the new endpoints (additive — old calls keep working meanwhile).	Token audit passes (dev-ui-ux-designer cross-platform consistency check); desktop screenshots approved by Ralph.
dev-mobile-rn	Ignition theme.js ; bottom-tab layout + 44px targets (sanctioned deviations); phone dashboard (KPI carousel + queue) and workspace (drawer client list) per Reports 01 §12/ §14 and 03 §8; offline counter notes per Report 05 §7.	Expo build ships both screens; token audit passes; sync_queue_depth telemetry live.
dev-qa-security	Full security scan per CLAUDE.md workflow (auth coverage, headers, CORS, error handling, rate limits); verify week-6 success metrics (Report 05 §8 table).	Gate B met. Metrics at target (T2 ≤35s, dismissals ≤15%, screens/session –40% ...).
all + Ralph	Only after Gate B: propose retirement of contacts-view.js / customer-aging-view.js and pruning of cockpit.css — file deletion needs Ralph's explicit approval per Safety Rules. Remaining 68-screen migration is a separate future package.	Ralph's written approval recorded before any deletion; rollback path retained until then.

Acceptance Checklist — Dashboard

Every line is testable and traceable to a source report. QA (dev-qa-security) signs each box; the screen does not ship with an open box.

Visual (Reports 02, 03)

- [] Canvas is **#0a0a0b**; cards **#131316**; popovers/hover **#1b1b1f** — verified by computed style, not screenshot.
- [] **#f97316** appears only as: primary CTA, active nav rail (2px), focus ring, links, primary chart series. No second accent hue anywhere.
- [] Cards are borderless (shadow + surface shift only), radius 16px; status pills and badges are the only sharp geometry.
- [] Headings render Plus Jakarta Sans (−0.04em); body Inter; **every money/quantity/aging figure renders JetBrains Mono with tabular-nums** (MoneyText component).
- [] Icon rail is 72px wide, icons 20px at 1.75px-equivalent stroke, active item carries orange rail + filled state.
- [] Layout matches Report 02 Fig A regions: KPI band → command bar → trend + Action Queue → insight panels; light mode via `[data-ig-theme="light"]` renders per Report 02 §10.
- [] All 68 legacy screens render pixel-identical to the Phase-0 baseline (sample of 10 compared).

Functional (Reports 04, 05)

- [] View mounts with **exactly one** call to `GET /api/dashboard/summary` (plus the two pre-existing reference-data calls); no `api.list("/api/parts")` full-catalog fetch remains.
- [] Action Queue items show severity × money-at-stake ordering; each deep-links to the entity-level action (e.g. payment sheet for that client), not just a screen.
- [] Queue lifecycle persists: acted/resolved/dismissed states survive refresh and re-login; dismissed items do not reappear for the same underlying signal state.
- [] `Ctrl + K` opens the palette on every screen; verbs `inv`, `pay`, `part`, `q`, `c`, `cust +`, `wa`, `#NNNN`, `> screen` all resolve; `part` answers inline without navigation; Enter executes top match.
- [] Live-updates poll runs at 30–60s with `If-None-Match`; unchanged data returns **304** with no body; a posted payment appears on the next poll (invalidation hook proven).
- [] Barcode wedge scan in palette or POS field resolves barcode → internal code → OEM code.
- [] Telemetry events fire: `task_start/complete`, `screen_view`, `palette_used`, `queue_item_*` with tenantId, role, taskId, ts.

Performance budgets (Reports 03 §10, 04 §2)

- [] Server p95: **≤350ms** cache-miss, **≤12ms** ETag 304 (load test at realistic tenant data volume).
- [] LCP **<1.8s** (KpiBand); TBT **<150ms**; skeleton crossfade 160ms, skipped when data lands ≤150ms.
- [] Total new Ignition CSS **<25KB gzipped**; palette keystroke-to-result **<50ms** on cached index.

Accessibility — WCAG 2.2 AA (Reports 01 §15, 03 §11)

- [] Contrast pairs measured: `#fafafa/#0a0a0b` ≥19:1; `#a1a1aa/#131316` ≥7:1; `#0a0a0b` on `#f97316` CTA ≥7:1; white-on-danger restricted to bold ≥13px labels.
- [] Full keyboard path: rail → command bar → KPI band → queue → panels in documented focus order; visible focus ring on every interactive element; palette close returns focus to trigger.
- [] Landmarks/roles per spec (nav, main, complementary; queue as list with actionable items); `prefers-reduced-motion` disables crossfades/slides; touch targets ≥24px desktop, 44px mobile.

Security (Report 04 §9; CLAUDE.md focus areas)

- [] Unauthenticated `GET /api/dashboard/summary` → 401; controller carries class-level `[Authorize]` (architecture test).
- [] Response DTO contains no `tenantId`, `dbName`, cache key, connection info or stack trace (the illustrative `cacheMeta` from Report 04's examples is stripped).
- [] Forced server error returns production-safe ProblemDetails; no .NET exception details reach the client.

Acceptance Checklist — Client Workspace & Cross-Cutting

Visual (Reports 02 Fig B, 03 §7)

- [] Three-region layout: searchable ClientRail (left) → ClientHeader with balance + AgingBar → TabStrip (Overview, Invoices, Payments, Quotes, Requests, Timeline/Messages); aging buckets use semantic colors (current→green ... 90+→danger).
- [] Narrow layout per Reports 01 §14 / 03 §8: rail becomes full-screen drawer; selected client pushes header+tabs as a page with back navigation.

Functional (Reports 04 §5–6, 05 §5)

- [] Selecting a client issues **one** `GET /api/clients/{id}/workspace` for header + balance + aging + KPIs; tabs lazy-load their paged child route only when opened.
- [] Typeahead uses `GET /api/clients/search`; results in ≤300ms; recents shown on empty query.
- [] **Receive payment without leaving the screen:** payment sheet opens from header/queue deep-link, amount allocates oldest-first, posts, and balance + aging + timeline update in place (T4 flow).
- [] New invoice / new quote start from workspace context with the client pre-attached (no customer dropdown hunt); quote→invoice conversion preserves context (Flow 05: 8→4 steps).
- [] WhatsApp reminder shows an editable **preview before send** (reuses `accountBalanceReminder` payload); delivery status lands back on the client's timeline, and failures create a queue item.
- [] Collection worklist: from the overdue queue item, the batch flow (5 clients) completes in ≤4 steps per client with notes captured (Flow 04 target).

Performance

- [] Workspace endpoint cached 20s per tenant+client; invalidated on payment/invoice mutation for that client; child tabs paged (default page size per contract, never unbounded).
- [] Rail search, tab switches and payment-sheet open all <200ms perceived (skeleton rules from Report 03 §9 apply).

Accessibility

- [] ClientRail: `role="listbox"`, options carry `aria-selected`, search input has an `sr-only` label; TabStrip follows the ARIA tabs pattern with arrow-key navigation.
- [] Payment sheet and drawers trap focus while open and return focus on close; all money values readable by screen reader with currency announced.

Security (Report 04 §5/§9)

- [] Cross-tenant customer id returns the **same 404** (status, shape, timing) as a nonexistent id — verified by integration test with two tenants.
- [] Ownership check runs before any child query (not relying on SQL predicate alone); all child routes carry `[Authorize]` and tenant filters.
- [] `/api/clients/search` rate limiting is partitioned per tenant; parameterized queries only (dev-database review signed).

Cross-cutting — package level

- [] **Zero legacy breakage:** no existing route renamed/removed/reshaped; WPF and mobile function unchanged against the old endpoints (smoke test both apps).
- [] **Week-6 success metrics** (Report 05 §8): T2 p50 ≤35s · T4 p50 ≤45s · T6 answered in-place ≥80% · palette-initiated sales ≥60% · queue same-day resolution ≥70% · dismissals ≤15% · screens/session −40% · reminder failures resolved ≤24h ≥90% · overdue>30d balance −20% by day 60.
- [] **Rollback armed:** default flips back to legacy automatically-decidable criteria (T2 p50 >60s or dismissals >30% at week 4) documented and rehearsed.
- [] **Cross-platform token audit** (dev-ui-ux-designer): web CSS vars, WPF ResourceDictionary and RN theme.js agree on every Report 02 §13 row.
- [] `dotnet build` + `dotnet test` (Architecture, Management, Integration suites) green after every phase; security scan workflow completed before Gate B.

Risk Register

Top 10 risks to the package, with likelihood / impact and the mitigation already designed into the reports (plus who watches it).

#	RISK	LIKELIHOOD	IMPACT	MITIGATION (AND OWNER)
R1	Legacy CSS collision — the 17,400-line <code>styles.css</code> (+392-line <code>cockpit.css</code>) breaks, or is broken by, Ignition styles across the other 68–70 screens	Medium	High	All Ignition rules scoped under <code>.ignition-root</code> , linked after legacy CSS, zero global resets; legacy files untouched; Phase-0 pixel baseline + 10-screen regression check each phase. (<i>dev-web-react</i> , <i>dev-qa-security</i>)
R2	No-build runtime limits — no JSX/bundler; hand-written <code>h()</code> components slow the team or drift from spec	Medium	Medium	Report 03 specs every component's props/states; mirrors existing <code>ui/</code> conventions; one component per file (workspace rule); composition maps prevent structural drift. (<i>dev-web-react</i>)
R3	WPF/mobile breakage via endpoint change	Low	High	Additive-only rule: 8 new routes, zero renames/removals/reshapes; old routes stay supported (composed, not forked); platforms migrate on their own schedule. Smoke-test both apps each backend release. (<i>dev-backend-api</i> , <i>dev-qa-security</i>)
R4	Stale money on screen — cache shows wrong receivables/cash after a write	Medium	High	Write-path invalidation hooks on sale/payment commits (the flagged medium-risk change), short TTLs (45s/20s), ETag revalidation; integration test: post payment → next poll updated. (<i>dev-backend-api</i>)
R5	Adoption resistance — counter staff keep the old screens; muscle memory beats the palette	Medium	Medium	4-week side-by-side (opt-in first), legacy reachable throughout, one-card training + self-teaching palette, defaults flip only at week 3–4, rollback criteria pre-agreed. (<i>workflow agent</i> , <i>Ralph</i>)
R6	Action-queue noise — items people dismiss erode trust in the whole dashboard	Medium	Medium	Precision guard: dismissals ≤ 15% is a shipping gate; lifecycle telemetry per item type; scoring (severity × money) tunable server-side without client release. (<i>dev-backend-api</i> , <i>workflow agent</i>)
R7	Projections don't materialize — the ~55% / ~98-min figures are planning assumptions (Report 05 itself models 57–98 min/day depending on method)	Medium	Low	"No projection ships unmeasured": telemetry replaces assumed frequencies from week 1; gates use measured p50s, not the projections; honest rollback either way. (<i>dev-qa-security owns the numbers</i>)
R8	Index migration on production DB — locking/perf during creation on large Transactions table	Low	Medium	Additive index-only migration, deployable ahead of code, run in low-traffic window; requires explicit approval per CLAUDE.md before touching production. (<i>dev-database</i>)
R9	Scale-out cache incoherence — in-memory per-tenant cache assumes the current single IIS/YARP host; a second API instance would serve stale/split views	Low	Medium	Explicitly documented assumption in Report 04 §04; trigger point defined: introduce a distributed cache (e.g. Redis) or sticky routing before any scale-out. (<i>dev-backend-api</i>)
R10	Third unreleased design language — Ignition joins Aurora and the live graphite-blue theme as another half-shipped look if scope creeps toward the other 68 screens	Medium	High	Hard scope: two screens only; Gate A locks the direction with Ralph before code; Gate B proves value before any expansion; 68-screen migration is explicitly a separate future package. (<i>documentation agent tracks scope</i> , <i>Ralph approves</i>)

Watch item (not yet a risk): screen-registry drift — 70 entries in Reports 01/03 vs 74 in Report 05. Recount at Phase-0 baseline so the "untouched screens" guarantee and regression sample are exact.

Open Questions for Ralph

The package deliberately leaves these to the owner. None block Phase 0; Q1 and Q2 are needed before Phase 1 ships. Each lists the default the team will assume if you simply say "use the defaults".

Q	DECISION NEEDED	CONTEXT	PACKAGE DEFAULT IF UNSPECIFIED
Q1	Dark or light as the shipped default? (per platform, with or without a user toggle)	Ignition is designed dark-first (Report 02 §2) but a complete light mode exists (§3, §10). Counter environments are bright; the "wow, it changed" effect is strongest in dark.	Dark default on web + WPF, per-user toggle via <code>[data-ig-theme]</code> ; mobile follows OS setting.
Q2	Approve the Ignition mockups (Gate A) — and flag anything to adjust before code	Design Change Workflow requires your sign-off on Report 02's hero figures (pages 8–12) before implementation starts.	No default — this one is blocking by design.
Q3	Platform order after web: WPF first or mobile first?	Phase 3 covers both; the roster can parallelize, but review bandwidth (your screenshots approval) serializes in practice.	WPF first (heaviest daily in-shop use), mobile immediately after.
Q4	LBP/USD display rules — default display currency, dual display, LBP rounding	Mockups show both <code>\$1,240.00</code> and <code>1,284,600 LBP</code> ; the currency-margin squeeze is an action-queue signal. Needs one rule: which currency leads KPIs, when to show both, and LBP rounding granularity.	USD leads KPIs; LBP secondary on hover/subline at current rate; LBP rounded to 1,000.
Q5	Who may dismiss Action Queue items?	Dismissal shapes what the shop pays attention to. Owner-only keeps control; staff-wide keeps the queue clean but risks hiding money issues.	Owner + accountant roles dismiss; counter staff can act and resolve but not dismiss.
Q6	Default landing screen per role	Dashboard-for-everyone is simplest; but counter staff (Karim) live in POS/Workspace, and Report 01's personas suggest role-based landing.	Dashboard for all in v1; revisit with week-6 <code>screen_view</code> telemetry.
Q7	Telemetry consent & retention	Report 05's beacons record per-user task timings (tenantId, role, taskId). It's your own staff's data, but you should set the retention window and who sees per-person numbers.	90-day retention; per-person data visible to owner only; aggregates for the team.
Q8	Naming: "Clients" or "Contacts"?	The new workspace replaces "Contacts"; the nav label and palette verb (<code>c</code>) should match what your staff actually say at the counter.	"Clients" (matches Report 04's <code>/api/clients/*</code> routes).
Q9	Legacy retirement (Gate B follow-through) — approve deleting <code>contacts-view.js</code> / <code>customer-aging-view.js</code> once metrics hit	Safety Rules require your explicit approval for any file deletion; until then they stay on disk, just unlinked from navigation.	No default — deletion always waits for your explicit "yes".

Sign-off block. Gate A (mockups approved, Q1 answered): _____ date _____ · Phase 1 shipped & after-screenshots approved: _____ · Phase 2 shipped: _____ · Gate B (metrics met, retirement approved): _____

Package manifest — docs/redesign/: 01-ux-research-user-flows · 02-visual-design-system · 03-ui-component-architecture · 04-system-architecture · 05-workflow-optimization · **06-consolidated-overview (this report)**. Each exists as .html (source of truth) and .pdf. Produced 2 July 2026 by the six-agent Ignition design team; no application code was modified in the making of this package.